Dylan Madisetti

Senior Software Engineer II, CoreWeave — LLM agent systems, evaluation harnesses, and scientific computing
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Summary

Senior software engineer and Mechanical Engineering PhD with 8+ years building software across industry and research. Owns core architecture for marimo, a 23,000+ star scientific notebook project, spanning reusable Python workflows, caching, and agent-facing correctness tooling. Background includes corresponding-author simulation research, NASA C++/MPI systems, Google production NLU, and auditable LLM agent harnesses.

Engineering Experience

CoreWeave 2025–present

Senior Software Engineer II — marimo (acquired Oct. 2025)

- Authored MEP 0008 and implemented reusable-notebook architecture spanning top-level-function serialization, setup-cell protections, and notebook embedding.
- Designed and implemented marimo's caching stack, including persistent and in-memory decorators, cache introspection, custom serialization, cell-level cached execution, and signed cache bundles.
- Shipped `marimo check`, correctness tooling built for humans and coding agents that emits machine-readable diagnostics; drove it from public design through configuration and production rules.
- Authored the specification and security boundaries and built a proof of concept for mo lab's sandboxed execution on CoreWeave Cloud; the product launched in public preview with GPU-backed sessions.
- Built a distributed-compute DAG prototype for marimo notebooks during a CoreWeave hackathon.
- Serve as a core maintainer of a 23,000+ star project (Nov. 2026): translate external user reports into shipped caching and pytest fixes, review contributors, publish releases, and mentor interns, junior colleagues, and SciPy participants.

Agent Systems & Evaluation

Cowboy 2026–present

Creator and maintainer

- Designed and open-sourced a Rust/Python/Nix runtime for persistent LLM agents with sandbox and controlled run-as boundaries, approval-gated effects, asynchronous message bridges, and audit-oriented execution records.
- Built and ran a prespecified four-task Harbor harness-validity calibration on Terminal-Bench 2.1 (1/4 task pass; 4/4 valid; zero infrastructure exclusions), validating a larger model-output cap and isolating a remaining runtime timeout.

Google — Mountain View Jul 2017–Jul 2018

Software Engineer, Natural Language Understanding

- Developed and evaluated debiasing methods for word, sentence, and semantic-vector representations on Google's NLU research team.
- Developed components of a multi-tower semantic-model framework and trained TensorFlow graphs for deployment at scale.
- Built massively parallel C++/MapReduce analysis pipelines and hyperparameter-autotuning procedures for black-box optimization.

NASA Langley Research Center Jun 2016–Aug 2016

Research and Development Intern

- Developed custom C++ Distributed Point Source Method simulation software for massively parallel MPI clusters.
- Built a laser/vibrometer system for guided-wave nondestructive evaluation of damaged composites and automated experimental equipment with C# serial I/O.

Earlier Software Experience

Software Developer, Equiscript and BoomTown (2012–2013); Co-founder / Software Developer, CloudCollege (2014–2015): shipped a beta-to-release product, ran user research, and helped secure $50,000 in seed funding.

Scientific Research

Johns Hopkins University 2018–2026

PhD Researcher; co-advised by Christopher D. Stiles, JHU APL

- Developed discrete-dislocation-dynamics and GPU-accelerated ray-tracing workflows to predict XRD peak broadening in additively manufactured 316L steel; corresponding author in *Journal of the Mechanics and Physics of Solids*.
- Built `pd3`, an open-source C++/Python parallel dislocation-dynamics framework, and applied ML to reconstruct dislocation structures from XRD and 3D simulations; presented results at MRS and TMS.
- Developed hyperspectral characterization analysis with JHU APL for in-situ monitoring of laser powder-bed-fusion builds.
- NSF Graduate Research Fellow and teaching assistant for Machine Learning for Mechanical Engineers.

Integrated Material Assessment & Predictive Simulation Laboratory, USC 2015–2017

Undergraduate Researcher

- Developed CAD discretization and CUDA-accelerated point-cloud meshing for distributed-point-source and peridynamic simulations; also researched energy harvesting and acousto-elastic metamaterials, leading to a publication.

Selected Publications and Talks

- Madisetti et al. "Computational investigation into XRD peak broadening effects with discrete dislocation dynamics." *JMPS* 196 (2025), corresponding author.
- Lavenstein et al. "The heterogeneity of persistent slip band nucleation and evolution in metals at the micrometer scale." *Science* 370 (2020).
- Madisetti et al. "Hash all the things: Caching for fast notebook restarts." *Proceedings of SciPy 2026*; presented "Python Notebooks Are Better Now" at PyAI 2026.

Technical Strengths

Languages & systems Python, C++, Rust, C#, SQL; Nix, WASM, MPI, CUDA, MapReduce, TensorFlow
Scientific computing Discrete dislocation dynamics, GPU ray tracing, XRD, DPSM, peridynamics, CAD meshing
Agent systems LLM harnesses, evaluations, sandboxing, approval gates, message bridges, distributed DAGs

Education

Johns Hopkins University PhD, Mechanical Engineering, conferred 2026; NSF GRFP
University of South Carolina BS, Mechanical Engineering, 2017; minors in Mathematics and Computer Science